

Lab 9 – Normalization Three

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April 15, 2024

1 Part 1

Yet again each primary key or set of keys functionally determines the rest of the columns. The primary keys in which the tables in the schema have these functional dependencies are as follows: for people it is the field id; for engineers it is the field pid; for flight control operators it is the field pid; for Astronauts it is the field pid; for crew it is the fields astronaut and spaceCraft; for space craft it is the field id; for systems it is the fields spaceCraft and name (this is done as it is assumed there is little to no value gained by having systems separate from space crafts as a system implemented in one spacecraft will have different parts or use than another space craft most of the time); for system compositions it is the fields spaceCraft systemName and part; for parts it is the field id; for catalog it is the fields supplier and part; for suppliers it is the field id.

2 Part 2

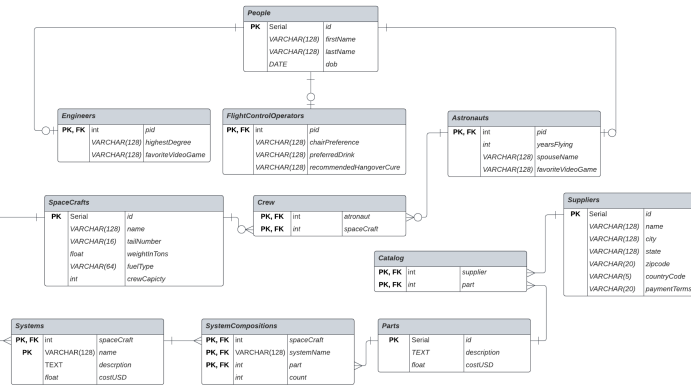


Figure 1: Schema

3 Part 3

The database is in 3NF Boyce-Codd Normal form because it is first in 1NF and 2NF. There is no multi-valued cells that contain multiple value per our business in a single cell (1NF). In every table, the non-key columns are dependent on the primary key (2NF). Every column is dependent only on the primary key columns and not any non-key columns (3NF). Every column is dependent on the whole unique primary key and not any one part of the primary key (BCNF).